Thought for 10s

I'll focus only on **equipment brochures/catalog PDFs** for non-ferrous metal extrusion lines — presses, handling systems, pullers, billet/log furnaces, die ovens, saws, aging ovens, and complete aluminum/copper extrusion plant equipment.

Thought for 13s

Yes — for **non-ferrous metal extrusion equipment brochures**, these are the better PDF catalogs/brochures to download. I removed product-profile catalogs and kept only **equipment / machinery** related PDFs.

#	Equipment brochure / catalog	Main equipment covered	PDF download link
1	**Belco — Extrusion Equipment Brochure**	Aluminum extrusion line equipment: furnaces, hot log saws, handling systems, die ovens, age ovens, pullers, washers, ovens, spray booths, finishing systems.	Download PDF
2	**Teyun Extrusion — Extrusion Catalogue**	Aluminum, magnesium, lead and alloy extrusion processing equipment. Includes extrusion press/line equipment for producing profiles, wire, tube, etc.	Download PDF
3	**SMS Group — Light Metal Extrusion Presses**	Light-metal extrusion presses for aluminum and aluminum alloys, including ancillary equipment and automation systems.	Download PDF
4	**UBE Machinery — SS-Hybrid Extrusion Press**	High-efficiency aluminum extrusion press; focuses on energy savings, operating efficiency, and modern press design.	Download PDF

5	**UBE Machinery — Company Brochure**	General machinery brochure including equipment for steel and non-ferrous metal manufacturing. Not only extrusion, but useful for UBE's machinery background.	Download PDF
6	**IAS / SMS Elotherm — ExtruLine**	Induction heating equipment for extrusion lines; useful for billet heating and thermal processing in ferrous and non-ferrous metal lines.	Download PDF
7	**Specialised Metal Extrusion Press Brochure**	Aluminum extrusion press, die preheating oven, material handling system, finish cut saw, hot log shear furnace, melting furnace, billet casting, extrusion puller, aging furnace.	Download PDF
8	**OMAV Reference List**	Reference list for aluminum extrusion lines: complete extrusion lines, double pullers, die ovens, aging ovens, basket handling systems. This is more of a reference/project list than a product brochure.	Download PDF

Best ones to start with:

1. **Belco** — best general aluminum extrusion equipment brochure.
2. **Teyun** — good if you want full aluminum/copper extrusion line equipment from a China supplier.
3. **SMS Group** — best high-end European extrusion press brochure.
4. **UBE** — good for extrusion press technology, especially aluminum and copper press systems.

Below are **recent public price ranges** for the main equipment types shown in the non-ferrous/aluminum extrusion equipment brochures. Most serious suppliers like Belco, SMS, UBE, OMAV, etc. are **quote-based**, so these prices are mainly from public China/global machinery listings and should be used as **budget estimates**, not final purchase prices.

Equipment	Recent public price range	Typical note

Complete aluminum extrusion production line	**US$185,000–1,000,000+**	Public listings for 650T–2000T aluminum extrusion lines commonly show **US$185k–1M**; some full-line listings show around **US$560k+ for 1150T**, **US$790k+ for 1600T**, and higher for 2000T lines.
Aluminum extrusion press, 650T–2000T	**US$80,000–1,000,000**	A 1000T press listing shows **US$80k–88k**, while broader 650T–2000T listings show **US$300k–1M** depending on tonnage, automation, and supplier.
2000T aluminum extrusion press	**~US$450,000–570,000+**	Teyun lists a 2000T press at **US$450k** with market price **US$500k**; other 2000T listings show **US$480k–570k**.
Billet heating furnace with hot log shear	**~US$60,000+**	Public listings show multi-billet aluminum heating ovens with hot log shear around **US$60k/piece**, but larger multi-log systems can be higher.
Die heating oven / die oven	**~US$10,000–15,000+**	Die ovens and extrusion die heating furnaces are often listed around **US$10k–15k**, depending on size and heating method.
Aging oven / aging furnace	**US$8,000–95,000**, sometimes higher	Made-in-China listings show aluminum extrusion aging ovens around **US$8k–95k**; Alibaba also shows examples starting around **US$45k** and **US$94k**.
Cooling table / run-out table / handling table	**US$35,000–55,000** for many systems; large systems can be higher	A Made-in-China cooling table listing shows **US$35k–45k**; another trading listing shows **US$35k–55k**. Full handling systems can reach **US$48k–200k** depending on length and included equipment.
Single puller / servo puller	**US$23,000–120,000**	Public aluminum extrusion puller listings show starting prices around **US$23k**, with used/factory puller machines around **US$45k–120k**.
Double puller	**US$40,000–200,000**	Double puller listings show common ranges around **US$40k–100k**, with some higher-end examples around **US$200k**.

Three-head / three-head puller	**US$180,000–280,000**	Public listings for aluminum extrusion three-head pullers show around **US$180k–280k**.
Stretcher	**Usually included in handling/cooling-table system; estimate US$30,000–100,000+**	Many listings bundle the stretcher with the cooling table/handling table, so standalone public pricing is less common. Handling-table systems that include stretcher and finish saw are listed around **US$48k–200k**.
Finish cut saw / product saw	**Usually bundled; estimate US$10,000–60,000+**	Finish saws are often included in the handling-table system, together with run-out table, stretcher, length gauge, and transfer table.
Induction billet heating / ExtruLine-type system	**Quote-based; often US$100,000–500,000+**	SMS Elotherm/IAS-type induction systems are industrial/high-end and normally quote-based. Public low-end furnace prices are not a good comparison for this level of equipment.
High-end SMS / UBE / OMAV / Belco full system	**Quote-based; likely US$1M–several million+**	These suppliers usually sell engineered extrusion systems, not commodity machines. Price depends heavily on press tonnage, automation, handling length, billet size, furnace type, quench system, saw automation, installation, and controls.

Simple estimate by factory size:

Factory / line type	Budget estimate
Small basic extrusion line, around 600T–800T	**US$185,000–400,000**
Medium line, around 1000T–1600T	**US$350,000–800,000**
Larger line, around 1800T–2200T	**US$500,000–1,200,000+**
European/Japanese/US engineered line	**US$1M–several million+**

For buying, I would ask suppliers for quotation using this minimum equipment list: **press tonnage, billet diameter, billet length, furnace type, hot log shear, cooling table length, puller type, stretcher, finish saw, aging oven capacity, die oven, quench system, automation level, installation, and warranty**.

give me the principle for the equipments
Thought for a second

Below is the **working principle** for the main non-ferrous metal extrusion equipment. I'll explain it in a practical factory-flow way, especially for **aluminum extrusion lines**, because most non-ferrous extrusion equipment brochures focus on aluminum.

1. Complete Aluminum / Non-Ferrous Extrusion Line

A complete extrusion line turns a heated metal billet into a long profile shape.

Basic principle:

1. Aluminum/copper/brass billet is heated.
2. The heated billet is loaded into the extrusion press.
3. A hydraulic ram pushes the billet through a die.
4. The die opening shapes the metal into the required profile.
5. The extruded profile comes out hot and soft.
6. It is pulled, cooled, stretched, cut, and aged to reach final strength.

Simple flow:

Billet heating → Extrusion press → Die → Run-out table → Cooling
→ Puller → Stretcher → Finish saw → Aging oven → Finished profile

The key idea is **plastic deformation**. The metal is not melted; it is heated until soft enough to be forced through the die.

---

2. Extrusion Press

The **extrusion press** is the heart of the line.

Working principle:

A heated billet is placed inside a container. A large hydraulic ram pushes the billet forward with very high force. The billet cannot move anywhere except through the die opening, so it flows through the die and becomes the die shape.

Hydraulic ram → Billet → Die → Extruded profile

For example, if the die hole is shaped like a tube, the output becomes a tube. If the die hole is shaped like a U-channel, the output becomes a U-channel.

Main parts:

Part	Function
Main cylinder / hydraulic ram	Pushes the billet
Container	Holds the billet during pressing
Die	Gives the final profile shape
Dummy block	Transfers ram force to billet
Die slide / die changer	Holds and changes dies
Hydraulic system	Creates pressing force

What it controls:

Parameter	Why it matters
Press tonnage	Maximum force available
Ram speed	Affects surface quality and production speed
Billet temperature	Affects flow and defects
Die temperature	Affects profile accuracy
Extrusion ratio	How much the billet cross-section is reduced

---

3. Billet Heating Furnace

The billet heating furnace heats the raw billet before extrusion.

Working principle:

The billet is heated to a controlled temperature so it becomes soft enough to extrude, but not melted.

For aluminum, billet temperature is commonly around **400–500°C**, depending on alloy and profile type. Copper/brass extrusion usually needs higher temperatures.

Types:

Type	Working principle
Gas billet furnace	Gas burners heat the furnace chamber and billet
Electric resistance furnace	Electric heaters heat the billet
Induction billet heater	Electromagnetic induction heats the billet internally

Why billet heating is important:

If the billet is too cold, extrusion force becomes too high and the profile may crack.
If the billet is too hot, the surface may become rough, weak, or defective.

---

4. Hot Log Shear / Hot Saw

This equipment cuts the heated long billet/log into the required billet length.

Working principle:

A long aluminum log is heated, then the hot log shear cuts it into shorter billets for the press. Some systems use a saw instead of a shear.

Long heated log → Hot shear/saw → Short billet → Press loader

Why it is used:

It allows the factory to heat one long log and cut only the needed billet length for each extrusion cycle. This improves material usage and production flexibility.

---

5. Die Heating Oven

The die heating oven preheats the extrusion die before it is installed in the press.

Working principle:

The die is heated to a controlled temperature before extrusion. This prevents the hot billet from contacting a cold die, which would cause poor metal flow, uneven profile shape, high stress, or die damage.

Why it matters:

A cold die can cause:

Problem	Result
Poor metal flow	Bad profile shape
High extrusion pressure	Press overload
Surface defects	Rough profile surface
Die thermal shock	Die cracking or shorter die life

The die oven makes the die temperature closer to the billet temperature so extrusion starts smoothly.

---

6. Die / Extrusion Tooling

The die is not always called “equipment,” but it is one of the most important parts.

Working principle:

The die is a very strong steel tool with a shaped opening. The metal flows through this opening and becomes the final profile.

There are two common die types:

Die type	Used for
Solid die	Solid profiles like bars, angles, channels
Hollow die / porthole die	Hollow profiles like tubes or box sections

For hollow aluminum profiles, the die splits the metal flow, forms the hollow shape around a mandrel, and then welds the metal flow back together under pressure.

---

7. Run-Out Table / Cooling Table

After leaving the press, the profile is very hot and soft. It lands on the run-out table.

Working principle:

The run-out table supports the long extruded profile as it exits the press. It moves the profile forward using rollers or belts. Cooling fans, air nozzles, water spray, or mist systems reduce the profile temperature.

Hot profile exits die → Supported by table → Cooled by air/water → Moved forward

Why it matters:

If cooling is uneven, the profile can bend, twist, or lose mechanical strength. For aluminum alloys like 6063 or 6061, cooling rate affects final hardness and strength.

---

8. Quenching System

Quenching is controlled rapid cooling after extrusion.

Working principle:

The hot profile is cooled quickly using air, water spray, water mist, or water bath. This locks the alloy structure in a condition suitable for later aging/hardening.

Common quenching types:

Type	Principle	Use
Air quench	Fans blow air onto profile	Thin/simple profiles
Water mist quench	Water mist cools faster than air	Medium cooling
Water spray quench	Water jets cool very fast	Stronger alloys/thicker profiles
Water bath quench	Profile passes through water	Heavy cooling requirement

Important point:

Quenching must be strong enough for mechanical properties but not so aggressive that it causes distortion.

---

9. Puller

The puller grabs the hot extrusion and pulls it along the run-out table.

Working principle:

As the profile exits the press, the puller clamps the front end and moves at a controlled speed. It keeps tension on the profile so it does not bend, twist, or pile up.

Profile exits press → Puller clamps front end → Puller moves with profile

Why it is important:

Without a puller, long profiles can bend, scratch, or become unstable. The puller helps keep the profile straight and controls movement after extrusion.

Types:

Puller type	Principle
Single puller	One puller clamps and pulls the profile
Double puller	Two pullers alternate, improving continuous production
Three-head puller	More advanced system for higher-speed automated lines

---

10. Double Puller

A double puller improves productivity compared with a single puller.

Working principle:

Two puller heads work alternately. While one puller is pulling the extruded profile, the other returns or prepares for the next cycle.

Puller A pulls profile
Puller B returns/prepares
Then Puller B takes over

Why it is better:

It reduces idle time and improves production continuity, especially for long profiles and faster extrusion lines.

---

11. Profile Saw / Flying Saw / Hot Saw

This saw cuts the extruded profile during or after extrusion.

Working principle:

The saw cuts the long extrusion into manageable lengths. Some saws cut the profile while it is still moving; these are often called flying saws. Others cut after the profile is stopped.

Types:

Saw type	Use
Hot saw	Cuts hot extruded profile near press/run-out area
Flying saw	Cuts moving profile without stopping production
Finish saw	Cuts final product to accurate customer length

Why it matters:

The first cut is usually rough. Final accurate cutting is done later by the finish cut saw.

---

12. Stretcher

The stretcher straightens the extruded profile after cooling.

Working principle:

The profile is clamped at both ends and pulled slightly in tension. This removes bending, twisting, and internal stress.

Clamp both ends → Pull slightly → Profile becomes straighter

Why it is needed:

Extruded profiles often come out slightly bent or twisted because of uneven cooling, die flow, or handling. Stretching improves straightness and dimensional stability.

Important point:

The stretcher does not reshape the profile heavily. It only applies controlled tension, usually a small percentage of elongation.

13. Finish Cut Saw

The finish cut saw cuts profiles to final customer length.

Working principle:

After stretching, the long profile is placed on a cutting table. The saw cuts it into exact required lengths, such as 3 m, 4 m, 6 m, or custom lengths.

Difference from hot saw:

Hot saw	Finish cut saw
Cuts profile during/soon after extrusion	Cuts after cooling and stretching
Rough production cut	Accurate final cut
Handles hot/soft material	Handles stable cooled profile

14. Aging Oven / Aging Furnace

The aging oven increases the strength and hardness of aluminum profiles.

Working principle:

After extrusion and quenching, some aluminum alloys are still not at final strength. The aging oven reheats the profiles at a lower controlled temperature for several hours. This causes precipitation hardening inside the aluminum alloy.

Quenched profile → Aging oven → Controlled heat treatment → Stronger profile

For many aluminum extrusion alloys, especially 6xxx series like **6061, 6063, 6082**, aging is very important.

Natural aging vs artificial aging:

Type	Principle
Natural aging	Strength increases slowly at room temperature

Artificial aging	Oven heats profiles to speed up strengthening

---

15. Induction Billet Heating System

Induction heating is a high-efficiency way to heat billets.

Working principle:

An alternating electromagnetic field is created around the billet. This induces electrical currents inside the metal. The electrical resistance of the billet turns those currents into heat.

Induction coil → Electromagnetic field → Eddy currents in billet → Heat

Advantages:

Advantage	Meaning
Fast heating	Short heating time
More efficient	Less wasted heat
Better control	More uniform billet temperature
Cleaner	No direct flame on billet

This is common in high-performance extrusion lines.

---

16. Billet Loader

The billet loader transfers heated billets into the extrusion press.

Working principle:

A mechanical arm, pusher, or conveyor moves the heated billet from the furnace/shear area into the press container.

Why it matters:

The billet is very hot, so manual handling is not practical. Automatic loading also reduces cycle time and improves safety.

---

17. Butt Shear

After extrusion, a small part of the billet remains in the press. This leftover part is called the butt.

Working principle:

The butt shear cuts off the remaining billet material after the useful extrusion is completed.

Extrusion finished → Billet butt remains → Butt shear removes it

Why butt is removed:

The butt often contains oxide, impurities, or poor-flow material, so it is usually not pushed through the die into the final product.

---

18. Handling System / Transfer System

The handling system moves profiles between cooling, stretching, cutting, aging, and packing areas.

Working principle:

Belts, rollers, chains, walking beams, or robotic transfer systems move profiles automatically through the line.

Why it matters:

Extruded profiles are long and easily scratched. A good handling system prevents damage and keeps production continuous.

---

19. Spray Booth / Finishing System

Some extrusion lines include finishing equipment after extrusion.

Working principle:

The profile surface is cleaned, treated, painted, powder-coated, or anodized depending on the required finish.

Common finishing methods:

Finish	Principle
Anodizing	Electrochemical oxidation creates hard protective oxide layer
Powder coating	Charged powder sticks to profile, then melts/cures in oven
Painting	Liquid coating applied and cured
Brushing/polishing	Mechanical surface finishing

This is not part of the extrusion press itself, but many complete aluminum extrusion factories include it.

---

20. Copper / Brass / Bronze Extrusion Equipment Principle

The general extrusion principle is similar to aluminum, but temperature, force, die design, and lubrication can be different.

Working principle:

Copper, brass, or bronze billet is heated and forced through a die using a hydraulic press.

Differences from aluminum extrusion:

Area	Aluminum	Copper/brass/bronze
Temperature	Lower	Usually higher
Force requirement	Medium	Often higher
Tool wear	Moderate	Can be higher
Lubrication	Sometimes less complex	Often more important
Products	Profiles, frames, tubes	Rods, bars, tubes, electrical/mechanical parts

Copper-alloy extrusion is often used for rods, tubes, busbars, electrical parts, bearings, bushings, and special industrial profiles.

Full Equipment Flow With Principles

1. Billet furnace
  Heats billet until soft enough for extrusion.

2. Hot log shear
  Cuts heated log into correct billet length.

3. Billet loader
  Loads billet into press container.

4. Extrusion press
  Pushes billet through die using hydraulic force.

5. Die
  Shapes metal into final cross-section.

6. Run-out table
  Supports long hot profile as it exits the press.

7. Quenching/cooling system
  Cools profile to control strength and shape.

8. Puller
  Pulls profile straight along table.

9. Hot/flying saw
  Cuts long extrusion into rough lengths.

10. Stretcher
  Straightens profile and removes internal stress.

11. Finish saw
  Cuts profile to final length.

12. Aging oven
  Heat-treats profile to final hardness/strength.

13. Finishing system
Adds anodizing, painting, powder coating, or surface treatment.

Most Important Equipment

If you are evaluating a supplier, the most important equipment are:

Priority	Equipment	Why important
1	**Extrusion press**	Determines maximum profile size and force
2	**Billet furnace**	Controls billet quality and energy cost
3	**Die oven / die system**	Controls profile accuracy and surface quality
4	**Run-out table + cooling/quench**	Controls straightness and mechanical properties
5	**Puller**	Controls profile stability and production speed
6	**Stretcher**	Corrects bending/twisting
7	**Aging oven**	Gives final strength/hardness
8	**Finish saw**	Controls final product length accuracy

then give me the technical datasheets for each equipment

I'll make **generic technical datasheets** for each equipment type, not supplier-specific certified datasheets, because exact values change by press size and manufacturer. I'll use typical industrial ranges and clearly mark what must be confirmed from the supplier quotation.

Thought for 8s

Below are **generic technical datasheets** for the main equipment in a **non-ferrous metal / aluminum extrusion line**. These are not one manufacturer's certified datasheets; they are practical procurement-style datasheets you can use to compare suppliers. Exact specs must be confirmed with the supplier based on alloy, billet size, press tonnage, profile size, automation level, and production capacity.

1. Complete Non-Ferrous / Aluminum Extrusion Line — Technical Datasheet

Item	Typical specification

Main use	Producing aluminum, copper, brass, bronze, or magnesium alloy profiles, rods, bars, tubes, and sections
Main process	Billet/log heating → extrusion press → die forming → run-out cooling → pulling → stretching → cutting → aging/heat treatment
Typical line capacity	Small line: 500T–800T press; medium: 1000T–1600T; large: 1800T–3000T+
Main products	Window/door profiles, industrial profiles, heat sinks, tubes, bars, busbars, rails, structural profiles
Main equipment included	Billet furnace, hot log shear/saw, billet loader, extrusion press, die oven, run-out table, quench/cooling system, puller, stretcher, finish saw, aging oven, handling/stacking system
Applicable materials	Aluminum alloys 1xxx/3xxx/6xxx/7xxx depending on press design; copper/brass/bronze require higher temperature and often higher force/tooling strength
Control system	PLC + HMI; optional SCADA/MES integration
Automation level	Manual, semi-automatic, automatic, or fully integrated line
Key buyer specs	Press tonnage, billet diameter, billet length, max profile width, table length, cooling method, puller type, aging oven capacity

The AEC extrusion manual lists the same major line equipment categories, including quenching system, log shear, puller, age/anneal ovens, handling equipment, saw/gauging system, stretcher, and stacking system.

---

2. Extrusion Press — Technical Datasheet

Item	Typical specification
Equipment name	Horizontal hydraulic extrusion press
Function	Forces heated billet through a die to form the final profile shape
Press type	Direct extrusion press; optional indirect extrusion for special applications

Typical tonnage	500T–6000T depending on product size; common aluminum lines use 650T, 800T, 1000T, 1250T, 1600T, 2000T
Common aluminum press range	650T–2000T for many small/medium profile factories
Billet diameter	Common aluminum billet sizes: 4 inch, 5 inch, 6 inch, 7 inch, 8 inch; larger presses use larger billets
Example billet size	One 1445MT listing uses 152 mm / 6 inch billet and 680 mm billet length
Extrusion speed	Often adjustable; example listing shows 0–13.1 mm/s ram speed
Main drive	Hydraulic system
Main components	Main cylinder, side cylinders, container, ram, dummy block, die slide, front platen, rear platen, hydraulic power unit
Control	PLC + HMI; manual/semi-auto/auto cycle
Main motor	Often tens to hundreds of kW depending on press size; example 1445MT listing shows 65 kW main motor
Key performance specs	Tonnage, billet size, container size, ram speed, profile exit speed, die stack size, hydraulic pressure, cycle time
Safety	Emergency stop, pressure relief, hydraulic interlocks, guard doors, temperature/pressure monitoring

Example public specs show a 1445MT aluminum extrusion press with 152 mm billet size, 680 mm billet length, 0–13.1 mm/s extrusion speed, 280 mm × 300 mm die stack, and 65 kW main motor.

---

3. Billet Heating Furnace — Technical Datasheet

Item	Typical specification
Equipment name	Aluminum billet heating furnace / log heating furnace
Function	Heats billets/logs to extrusion temperature before pressing
Heating type	Gas, electric resistance, or induction
Material	Aluminum billet/log; special designs for copper/brass/bronze

Aluminum working temperature	Commonly around 450–520°C for many aluminum extrusion systems
Billet diameter	Example public furnace range: 80–330 mm
Log length	Example public range: 6000–8000 mm
Furnace style	Single-billet furnace, multi-billet furnace, long-log furnace
Temperature control	Multi-zone temperature control recommended
Loading method	Conveyor, pusher, walking beam, or loader
Optional equipment	Hot log shear, hot saw, billet brush, billet transfer system
Main control	PLC + HMI
Key buyer specs	Billet diameter, billet length, heating capacity kg/h, fuel type, temperature uniformity, energy consumption

A public multi-billet furnace specification gives billet diameters from 80 mm to 330 mm and log lengths from 6000 mm to 8000 mm. A single billet furnace example lists 380 V / 50 Hz and working temperature of 450–520°C.

---

4. Hot Log Shear / Hot Saw — Technical Datasheet

Item	Typical specification
Equipment name	Hot log shear or hot log saw
Function	Cuts heated long logs into short billets before extrusion
Material	Aluminum logs; special blade/shear design for copper alloys
Cutting method	Hydraulic shear or circular saw
Installed position	Between billet furnace and press loader
Billet temperature	Must handle hot billet/log temperature
Cut length	Adjustable billet length according to press recipe

Control	PLC-controlled cut length; often integrated with furnace and press
Main buyer specs	Max log diameter, max log length, cut length range, cut accuracy, cycle time, blade/shear life
Safety	Guarding, interlock, emergency stop, hot-material protection

Hot shear is usually part of a billet furnace/log heating system; suppliers often quote it together with the furnace and billet transfer system.

---

5. Die Heating Oven — Technical Datasheet

Item	Typical specification
Equipment name	Extrusion die oven / die preheating oven
Function	Preheats extrusion dies before installation in the press
Heating type	Electric resistance, gas, infrared, or induction-assisted
Typical die temperature	Depends on alloy; often near extrusion starting condition
Structure	Drawer type, trolley type, box type, or multi-chamber
Capacity	Single die, multiple dies, or die stack sets
Temperature control	PID temperature controller or PLC
Temperature uniformity	Important for die life and stable metal flow
Key buyer specs	Die diameter, die thickness, number of die pockets, heating time, temperature range, uniformity, power rating
Safety	Over-temperature alarm, door interlock, insulated chamber

Die heating is important because a cold die can cause poor metal flow, higher extrusion force, rough surface, and die thermal shock.

---

6. Extrusion Die / Tooling — Technical Datasheet

Item	Typical specification
Equipment name	Extrusion die / die stack / tooling
Function	Gives the profile its final cross-section
Die types	Solid die, semi-hollow die, hollow/porthole die
Material	Hot-work tool steel, commonly H13 or equivalent
Products	Flat bar, angle, channel, tube, hollow profile, heat sink, custom profile
Die stack components	Die, backer, bolster, feeder plate, mandrel/bridge for hollow profiles
Important design factors	Bearing length, flow balance, extrusion ratio, weld chamber, profile tolerance
Key buyer specs	Profile drawing, alloy, temper, tolerance, surface finish, annual production volume
Maintenance	Nitriding, polishing, correction, cleaning, storage
Failure risks	Cracking, wear, deflection, poor flow balance, die lines

For hollow aluminum profiles, the die splits the metal flow around a mandrel and welds it back together under pressure.

---

7. Run-Out Table / Cooling Table — Technical Datasheet

Item	Typical specification
Equipment name	Run-out table / cooling table / extrusion table
Function	Supports and transfers hot profiles after they exit the press
Structure	Steel frame with rollers, belts, graphite plates, or felt belts
Drive type	Idler rollers, motorized rollers, belts, chain-driven system
Cooling method	Air fan, duct cooling, water mist, water spray, or flood quench
Position	Immediately after press exit

Table length	Depends on max profile length and press size
Main buyer specs	Table length, profile width, max profile weight, belt/roller type, cooling capacity, line speed
Automation	Integrated with puller, saw, stretcher, and handling system
Safety	Guarding, emergency stop, hot-profile protection

Presezzi describes extrusion tables with strong steel structure, idle or motorized rolls, under-roll cooling using axial/centrifugal fans and ducts, movable/fixed top cooling, and automatic raising rollers. Belco also lists run-out equipment, over-table duct cooling, and under-table duct cooling in its extrusion equipment brochure.

---

8. Quenching / Cooling System — Technical Datasheet

Item	Typical specification
Equipment name	Profile quenching system / air-water cooling system
Function	Rapidly cools extruded profile to control mechanical properties and shape
Cooling media	Air, water mist, water spray, water bath, or combined air-water
Application	Especially important for heat-treatable aluminum alloys such as 6061, 6063, 6082
Control	Adjustable fan speed, water pressure, nozzle zones, recipe-based cooling
Position	On run-out table after die exit
Key buyer specs	Cooling rate, number of zones, fan capacity, water flow, water pressure, profile size range
Quality effect	Controls hardness, straightness, distortion, surface quality
Maintenance	Nozzle cleaning, water filtration, fan inspection, corrosion control

Cooling systems are commonly integrated with the run-out table and puller system. Granco Clark lists cooling and quench systems as part of extrusion processing equipment, including air quenching and medium cooling systems.

---

9. Single Puller — Technical Datasheet

Item	Typical specification
Equipment name	Single-head extrusion puller
Function	Clamps and pulls the hot profile as it exits the press
Drive type	Cable-driven, chain-driven, servo, hydraulic, or electric
Puller head	Clamp/gripper for profile front end
Speed control	Synchronized with extrusion speed
Position	Runs along puller track beside or above run-out table
Main buyer specs	Pulling force, max speed, track length, profile size range, clamp type
Benefits	Keeps profile straight, reduces bending, prevents pile-up, improves surface quality
Control	PLC integrated with press and saw
Safety	Anti-collision, emergency stop, clamp interlock

Belco states that its pullers range from lightweight cable-driven designs to heavy-duty chain-driven systems.

---

10. Double Puller / Twin Puller — Technical Datasheet

Item	Typical specification
Equipment name	Double puller / twin puller
Function	Two puller heads alternate pulling, allowing smoother continuous production
Puller type	Bypassing double-head, twin puller, or two independent pullers
Main advantage	Reduces idle time between extrusion cycles

Drive	Servo, chain, cable, or electric drive depending on supplier
Integration	Often integrated with flying saw and run-out table
Main buyer specs	Pulling force, head travel length, max speed, handoff/bypass method, synchronization accuracy
Best use	Medium/high-output aluminum extrusion lines
Control	PLC/HMI with automatic cycle logic
Safety	Collision avoidance, clamp monitoring, emergency stop

Belco describes its dual puller as a bypassing-type puller with no hand-off from one puller to the other. Kautec lists downstream options such as single puller with positioned saw, puller with flying saw, and two independent pullers.

---

11. Three-Head Puller — Technical Datasheet

Item	Typical specification
Equipment name	Three-head extrusion puller
Function	Advanced puller system for high-output automatic extrusion lines
Puller heads	Three coordinated pulling/cutting/returning heads
Main advantage	Higher productivity and reduced dead time
Best use	High-speed lines, long profiles, automated production
Control	Full PLC/servo synchronization
Integration	Press, run-out table, hot saw/flying saw, cooling system
Main buyer specs	Number of puller heads, pulling force, max speed, track length, synchronization method
Complexity	Higher than single/double puller
Maintenance	Servo/drive system, clamps, track alignment, sensors

This is usually chosen only when line output justifies higher automation cost.

---

12. Hot Saw / Flying Saw — Technical Datasheet

Item	Typical specification
Equipment name	Hot saw / flying hot extrusion saw
Function	Cuts the moving hot extrusion to rough production length
Cutting type	Stationary cut or flying cut synchronized with profile movement
Blade type	Circular saw blade suitable for hot aluminum/profile material
Position	On run-out side, often near puller system
Main buyer specs	Max profile width/height, blade diameter, cutting speed, cutting accuracy, moving-cut capability
Control	PLC/servo synchronized with puller and profile speed
Dust/chip handling	Chip collection and extraction recommended
Safety	Blade guarding, interlock, emergency stop

Belco's brochure lists flying hot extrusion saws as part of extrusion puller/run-out equipment.

---

13. Stretcher — Technical Datasheet

Item	Typical specification
Equipment name	Profile stretcher
Function	Straightens profiles and removes internal stress after cooling
Working method	Clamps both ends and applies controlled tensile stretch
Material	Aluminum profiles; sometimes copper alloy profiles depending on line
Stretch amount	Usually small controlled elongation, based on alloy/profile requirement
Clamp system	Hydraulic or mechanical clamps

Main buyer specs	Max profile length, max profile width, pulling force, clamp opening, elongation control
Quality effect	Improves straightness, reduces twist, improves dimensional stability
Control	Manual, semi-auto, or PLC-controlled
Safety	Clamp interlock, tension overload protection, emergency stop

Ecofill lists stretchers as one of the required aluminum extrusion industry equipment types, alongside presses, handling systems, hot saws, quench systems, finish saws, age ovens, die ovens, billet furnaces, log furnaces, and shears.

---

14. Finish Cut Saw — Technical Datasheet

Item	Typical specification
Equipment name	Finish saw / precision saw / final cut saw
Function	Cuts cooled and stretched profiles to final customer length
Cutting type	Precision circular saw
Material	Aluminum profiles; blade selected for alloy/profile wall thickness
Cut length	Common customer lengths: 3 m, 4 m, 6 m, custom
Accuracy	Depends on measuring system and clamping
Main buyer specs	Max profile bundle size, blade diameter, cut length range, length tolerance, cutting speed
Optional features	Automatic length gauge, bundle cutting, chip collection, conveyor loading/unloading
Safety	Blade guard, clamp interlock, dust/chip extraction

This is different from the hot/flying saw: the finish saw makes the accurate final cut after cooling and stretching.

---

15. Aging Oven / Aging Furnace — Technical Datasheet

Item	Typical specification
Equipment name	Aluminum profile aging oven / artificial aging furnace
Function	Heat-treats profiles to increase strength/hardness
Main process	Artificial aging / precipitation hardening
Applicable alloys	Heat-treatable aluminum alloys, especially 6xxx series like 6061, 6063, 6082
Heating type	Gas or electric
Typical temperature range	Supplier-dependent; often low-temperature heat treatment compared with billet heating
Structure	Batch oven, trolley oven, continuous oven
Air circulation	Strong recirculation fan for uniform temperature
Loading method	Basket, rack, trolley, or automatic handling
Main buyer specs	Oven size, loading capacity, temperature uniformity, heating time, max temperature, energy type
Control	PLC/PID temperature control, recipe storage, chart recorder optional
Quality effect	Controls final temper, hardness, tensile strength

Granco Clark lists age and anneal ovens among extrusion processing/off-line equipment.

---

16. Induction Billet Heating System — Technical Datasheet

Item	Typical specification
Equipment name	Induction billet heater / ExtruLine-type induction heating system
Function	Heats billets using electromagnetic induction
Heating principle	Eddy currents generated inside billet create heat
Material	Aluminum, copper, brass, and other conductive non-ferrous metals

Benefits	Fast heating, high efficiency, good temperature control, cleaner than gas flame
Main components	Induction power supply, induction coil, billet conveyor, cooling system, temperature sensors
Control	PLC/HMI, pyrometer feedback, recipe-based heating
Main buyer specs	Billet diameter, billet length, throughput kg/h, power kW, temperature uniformity, energy consumption
Cooling	Coil and power supply require water cooling
Best use	High-efficiency, high-output extrusion lines

SMS Elotherm/IAS positions ExtruLine as induction heating equipment for extrusion lines.

---

17. Billet Loader / Furnace-to-Press Transfer — Technical Datasheet

Item	Typical specification
Equipment name	Billet loader / billet transfer system
Function	Transfers heated billet from furnace/shear to press container
Transfer type	Roller conveyor, pusher, robotic arm, loader arm, walking beam
Material	Hot aluminum billet/log
Integration	Furnace, hot shear, press, container
Main buyer specs	Billet diameter, billet length, billet weight, transfer speed, cycle time, alignment accuracy
Control	PLC interlocked with press cycle
Safety	Hot-material guarding, emergency stop, position sensors
Quality effect	Reduces billet temperature loss and cycle time

Granco Clark lists furnace-to-press billet transfer systems as extrusion equipment.

---

18. Butt Shear — Technical Datasheet

Item	Typical specification
Equipment name	Butt shear
Function	Cuts/removes leftover billet butt after extrusion
Position	At press exit/container area
Drive	Hydraulic shear
Material	Hot aluminum/copper alloy butt
Purpose	Prevents oxide/impurity-rich butt from entering finished profile
Main buyer specs	Shear force, billet diameter range, cycle time, blade material, integration with press
Control	Interlocked with press cycle
Maintenance	Blade inspection, hydraulic system, alignment

The butt is usually discarded or recycled because it may contain oxide, poor-flow material, or contamination.

---

19. Handling / Transfer / Stacking System — Technical Datasheet

Item	Typical specification
Equipment name	Profile handling system / transfer table / stacking system
Function	Moves profiles between cooling, stretching, cutting, aging, and packing
Transfer method	Belt, roller, chain, walking beam, lift table, robotic transfer
Material	Long aluminum/non-ferrous profiles
Surface protection	Felt belts, graphite, low-friction rollers, plastic/nylon supports
Automation	Manual, semi-auto, or automatic stacking

Main buyer specs	Profile length, bundle weight, transfer speed, surface protection, automation level
Quality effect	Reduces scratches, bending, collision damage
Control	PLC + sensors
Safety	Guarding, emergency stops, anti-collision sensors

The AEC manual includes handling equipment and stacking systems as part of the extrusion line equipment set.

---

20. Surface Finishing System — Technical Datasheet

Item	Typical specification
Equipment name	Anodizing / powder coating / painting / polishing line
Function	Adds corrosion protection, color, surface hardness, or decorative finish
Common finishes	Anodizing, powder coating, painting, brushing, polishing
Main process examples	Cleaning → pretreatment → coating/anodizing → curing/sealing → inspection
Materials	Mainly aluminum profiles; copper/brass use different finishing processes
Main buyer specs	Profile size, line speed, coating thickness, color system, oven size, pretreatment chemistry
Quality checks	Coating thickness, adhesion, corrosion resistance, color consistency
Environmental needs	Wastewater treatment, exhaust/filtration, chemical handling
Safety	Chemical safety, ventilation, fire protection for powder coating

This is downstream of extrusion, but many complete extrusion factories include finishing lines.

Supplier RFQ Checklist

When you ask a supplier for a real datasheet/quotation, ask them to fill these fields:

Required field	Why it matters
Press tonnage	Determines profile size and extrusion force
Billet diameter and length	Must match furnace, loader, container, and press
Alloy range	Aluminum vs copper/brass/bronze changes temperature and force
Max profile width/height	Determines die and press capability
Annual/monthly production target	Determines automation level
Required profile length	Determines table, puller, stretcher, and saw size
Cooling method	Controls strength and distortion
Puller type	Single/double/three-head affects productivity
Aging oven capacity	Must match line output
Control system brand	Affects maintenance and spare parts
Power/fuel requirement	Needed for factory utility planning
Installation/training/warranty	Important for total project cost

For a basic aluminum extrusion factory, the **minimum core equipment** is: **billet furnace, hot shear/saw, extrusion press, die oven, run-out/cooling table, puller, stretcher, finish saw, aging oven, and handling/stacking system**.